DATA ANALYTICS DIVISION · INTERNAL TECHNICAL PUBLICATION

Nifty 100 Financial Intelligence Platform

Comprehensive Analyst User Guide & System Architecture Specification

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Effective Date:	November 2026
Target Audience:	Equity Research Analysts, Portfolio Managers, Risk Auditors
Coverage Universe:	92 Nifty 100 Index Constituents (10–14 Year Historical Time Series)
Primary Engine Modules:	ETL Lake, 50+ KPI Engine, 6 Screeners, ML Clustering, FastAPI, Streamlit
Technical Platform:	Python 3.10, SQLite 3, FastAPI, Streamlit, Plotly, ReportLab

Executive Notice: This technical manual serves as the standard operating reference for the Nifty 100 Financial Intelligence Platform. All data transformations, financial ratios, valuation indicators, and machine learning clustering models described herein adhere strictly to reproducible quantitative accounting standards. Unauthorized redistribution is strictly prohibited.

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Platform 7-Layer Architecture

The platform is engineered as a robust, single-file serverless relational stack decoupled into seven logical layers, guaranteeing complete reproducibility and sub-second analytical querying.

Layer	Layer Name	Components & Primary Technologies
L1	Raw Ingestion	openpyxl, Excel headers normalization, requests URL validator
L2	ETL & Validation	Ticker normaliser, YYYY-MM year converter, 16 DQ validation rules
L3	Persistent Storage	SQLite 3.x (data/nifty100.db), foreign key constraint enforcement
L4	Analytics Core	Vectorized KPI engine, turnaround CAGR logic, leverage & cash flow
L5	Intelligence & ML	KMeans k=5 clustering, StandardScaler, NLP rule engine (pros/cons)
L6	Reporting Layer	ReportLab Platypus tearsheets (92 PDFs), sector summaries, Excel
L7	User Interfaces	Streamlit multi-page dashboard (8 screens), FastAPI (16 endpoints)

2. Dataset Catalogue & Master Entity Data Model

The data warehouse unifies 7 core datasets extracted from statutory filings and 5 curated supplementary reference datasets. The primary entity key is the National Stock Exchange (NSE) trading ticker (e.g., TCS, HDFCBANK).

Dataset File	Target DB Table	Row Count	Coverage & Nature
companies.xlsx	companies	92	Master entity reference, ISIN, corporate description, profile URLs
profitandloss.xlsx	profitandloss	1,070	Annual P&L; history (FY10–FY24), Revenue, EBITDA, PAT, EPS
balancesheet.xlsx	balancesheet	1,140	Annual Balance Sheet, Equity, Borrowings, Net Worth, Total Assets
cashflow.xlsx	cashflow	1,056	Statutory Cash Flow statements: Operating (CFO), Investing (CFI), Financing (CFF)
documents.xlsx	documents	1,456	BSE India official annual report direct PDF repository URLs
prosandcons.xlsx	prosandcons	14	Qualitative investment commentary & corporate highlights
sectors.xlsx	sectors	92	11 Macro sectors & 33 granular sub-industry taxonomy mappings
stock_prices.xlsx	stock_prices	5,520	60-month historical OHLCV price series (Jan 2020 – Dec 2024)
market_cap.xlsx	market_cap	552	Annual valuation multiples (P/E, P/B, EV/EBITDA, Dividend Yield)
peer_groups.xlsx	peer_groups	56	11 dedicated peer clusters with benchmark anchor designations
financial_ratios.xlsx	financial_ratios	1,155	Computed analytical ratios, CAGR growth rates, and health metrics

Relational Integrity & Join Standards

Primary Foreign Key Rules: All subsidiary tables enforce strict foreign key constraints referencing companies.id. Composite natural keys (company_id, year) are utilized for all longitudinal financial statement tables. All queries combine statements via USING (company_id, year).

3. Financial Ratio Engine: 50+ KPIs & Edge Cases

The Ratio Engine (src/analytics/ratio_engine.py) computes core profitability, return, leverage, efficiency, cash quality, and growth metrics. It rigorously handles accounting boundary conditions and edge cases:

Metric Name	Mathematical Formula	Institutional Benchmark	Edge-Case Execution Rule
Net Profit Margin	PAT / Revenue × 100	> 15% (Healthy)	Returns None if Revenue = 0; allows negative margins.
Operating Margin (OPM)	Operating Profit / Revenue × 100	> 20% (Strong)	Cross-validated against source opm_percentage within ±1%.
Return on Equity (ROE)	PAT / (Equity + Reserves) × 100	> 15% (Compounding)	**CRITICAL:** Returns None if Total Equity ≤ 0 (avoids inverted signs).
Return on Capital (ROCE)	EBIT / (Net Worth + Borrowings) × 100	> 18% (Superior)	Banks/NBFCs evaluated via sector-relative benchmark.
Debt-to-Equity (D/E)	Borrowings / Total Equity	< 1.0x (< 0.5x Conservative)	0.0 for debt-free companies; Financials carved out in screener.
Interest Coverage (ICR)	(Operating Profit + Other Inc) / Interest	> 3.0x (Safe)	If Interest = 0, returns None and displays 'Debt Free' (not Inf).
Free Cash Flow (FCF)	CFO + CFI	> 0 (Self-funding)	Negative allowed; 3 consecutive negative years triggers distress alert.
CFO / PAT Quality Ratio	Operating Cash Flow / PAT	> 1.0x (Cash-backed)	< 0.5x flags 'Accrual Risk'; None if PAT = 0.
CapEx Intensity	abs(CFI) / Sales × 100	< 3% Light, > 8% Heavy	Proxy using net investing cash outflows.
Asset Turnover	Sales / Total Assets	> 1.0x (Capital efficient)	Returns None if Total Assets = 0.

CAGR Boundary Decision Matrix

Compound Annual Growth Rates (3Y, 5Y, 10Y) for Sales, PAT, and EPS strictly adhere to Section 23.1 rules: If base year is negative and final year is positive, the engine sets CAGR = None and assigns the **TURNAROUND** flag. If base year is positive and final year is negative, it returns **DECLINE_TO_LOSS**. If base year is zero, it returns **ZERO_BASE**. Under no circumstances are imaginary roots generated.

4. Quantitative Investment Screener & Presets Reference

The Screener Engine (src/screener/engine.py) exposes 15+ interactive filters and 6 institutional preset screening strategies. All filtering thresholds are externalized in config/screener_config.yaml.

Preset Strategy	Screening Filter Logic	Ranking KPI	Expected Count
Quality Compounder	ROE > 15%, D/E < 1.0, FCF > 0, Sales CAGR 5Y > 10%	Composite Score (Desc)	15 – 35 Cos (38 Actual)
Value Pick	P/E < 20x, P/B < 3.0x, D/E < 2.0x, Dividend Yield > 1.0%	FCF Yield (Desc)	10 – 25 Companies
Growth Accelerator	PAT CAGR 5Y > 20%, Sales CAGR 5Y > 15%, D/E < 2.0x	PAT CAGR 5Y (Desc)	8 – 20 Companies
Dividend Champion	Dividend Yield > 2.0%, Dividend Payout < 80%, FCF > 0	Dividend Yield (Desc)	10 – 20 Companies
Debt-Free Blue Chip	Borrowings = 0 (or D/E = 0), ROE > 12%, Revenue > ■5,000 Cr	ROE (Desc)	15 – 30 Companies
Turnaround Watch	Sales CAGR 3Y > 10%, FCF positive in latest year, D/E declining	Sales CAGR 3Y (Desc)	5 – 15 Companies

Screener Workflow & Operational Instructions

1. **Web Interface Execution:** Navigate to Screen 03 (pages/03_screener.py) on localhost:8501. Click any preset button to instantly configure the multi-slider filters. The live results table re-computes dynamically.
2. **Export Capability:** Click the Download CSV button below the filtered table to export a standardized CSV containing all 14 core metrics for offline modeling.
3. **Programmatic API Querying:** Execute HTTP GET requests to /api/v1/screener?min_roe=15&max;_de=1.0 for real-time JSON result arrays.

5. Financial Health & Composite Scoring Model

To enable equitable cross-sector comparisons, the platform implements a **Sector-Relative Winsorised Scoring Model**. Raw metrics are grouped by macro-sector, winsorised at the 10th and 90th percentiles to neutralize extreme outliers, and scaled into a standard 0 to 100 distribution.

Analytical Dimension	Weight	Sub-KPI Breakdown	Mathematical Normalisation
Profitability & Efficiency	**50%**	ROE (33.3%), ROCE (33.3%), Net Profit Margin (33.4%)	Higher is better. Winsorised at P10/P90 within sector.
Growth Momentum	**30%**	Revenue CAGR 5Y (50%), PAT CAGR 5Y (50%)	Higher is better. Turnaround or loss flags assigned score = 0.
Valuation Attractiveness	**20%**	Price-to-Earnings (50%), Price-to-Book (50%)	Lower is better (inverse scaling: (P90 - val) / (P90 - P10) × 100).

Composite Quality Score Triage Bands

The resulting Composite Quality Score (0–100) provides instantaneous portfolio triage:

- **Tier 1: Elite Compounders (Score ≥ 75):** High capital returns backed by secular growth and disciplined balance sheets.
- **Tier 2: Quality Core (60 ≤ Score < 75):** Robust businesses with steady profitability and moderate leverage.
- **Tier 3: Moderate / Neutral (45 ≤ Score < 60):** Cyclical or capital-intensive operations sensitive to macro trends.
- **Tier 4: Watchlist / Stressed (Score < 45):** Low return on capital, declining growth trajectories, or elevated balance sheet gearing.

6. Peer Comparison Engine & Radar Benchmarking

The Peer Engine (src/analytics/peer.py) groups companies into 11 competitive sub-industry peer groups designated in data/supporting/peer_groups.xlsx. For every company and financial metric, it computes intra-peer percentile ranks via PERCENT_RANK() OVER (PARTITION BY peer_group ORDER BY metric).

Peer Group Cluster	Constituent Member Companies	Anchor Benchmark
Private Banks	HDFCBANK, ICICIBANK, AXISBANK, KOTAKBANK, INDUSINDBK	HDFCBANK
Public Banks	SBIN, BANKBARODA, CANBK, PNB	SBIN
IT Services	TCS, INFY, HCLTECH, TECHM, LTIM	TCS
Pharmaceuticals	SUNPHARMA, CIPLA, DRREDDY, DIVISLAB, TORNTPHARM	SUNPHARMA
Automobiles	MARUTI, TATAMOTORS, M&M;, BAJAJ-AUTO, EICHERMOT, HEROMOTOCO, TVSMOTOR	MARUTI
Life Insurance	LICI, HDFCLIFE, SBILIFE, ICICIPRULI	LICI
Oil & Gas	RELIANCE, ONGC, BPCL, IOC, GAIL	RELIANCE
Power & Utilities	NTPC, POWERGRID, TATAPOWER, ADANIPOWER, NHPC, JSWENERGY, ADANIGREEN	NTPC
Steel & Metals	TATASTEEL, JSWSTEEL, JINDALSTEL, HINDALCO	TATASTEEL
FMCG / Staples	HINDUNILVR, ITC, BRITANNIA, DABUR, NESTLEIND, GODREJCP, TATACONSUM	HINDUNILVR
Consumer Finance	BAJFINANCE, CHOLAFIN, SHRIRAMFIN	BAJFINANCE

8-Axis Radar Chart Interpretation

Radar charts (saved in reports/radar_charts/ and accessible on Dashboard Screen 04) evaluate: **ROE, ROCE, NPM, D/E (inverted), FCF, PAT CAGR 5Y, Revenue CAGR 5Y, and EPS CAGR 5Y**. A polygon expanding toward the outer perimeter indicates superior competitive performance against the peer group average (dashed benchmark circle).

7. Cash Flow Intelligence & Capital Allocation

Accounting profits can be distorted by non-cash accruals. The Cash Flow Intelligence module (src/analytics/cashflow_kpis.py) dissects core cash generation and classifies every company into one of 8 capital allocation sign patterns.

Sign Pattern (CFO, CFI, CFF)	Classification Label	Economic Rationale & Business Meaning
(+ , – , –)	Reinvestor / Debt Repayer	Ideal compounding profile: Organic operations fund both CapEx and debt retirement / dividends.
(+ , + , –)	Asset Seller / Debt Repayer	Downsizing or non-core asset divestment utilized to repay borrowings.
(+ , – , +)	Reinvestor / Raising Capital	Aggressive capacity expansion: Operating cash supplemented with external debt/equity.
(+ , + , +)	Asset Seller / Raising Cap	Cash hoarding / strategic liquidity buildup across all financing channels.
(– , – , –)	Distress Burning Cash	High risk: Cash reserves rapidly depleting across all operational and investing fronts.
(– , + , –)	Distress Asset Seller	Operational losses covered by divesting productive fixed assets to service liabilities.
(– , – , +)	Distress Raising Capital	External borrowing or dilution strictly to fund ongoing operational losses (Red Flag).
(– , + , +)	Distress Raising / Selling	Severe cash burn requiring emergency asset sales and debt capital infusions.

Early Distress Warning Signals

The system triggers an automated distress alert in output/distress_alerts.csv whenever a company demonstrates:

1. **Operating Cash Burn:** CFO < 0 in conjunction with positive financing flows (CFF > 0).
2. **Sustained Negative Free Cash Flow:** FCF negative for 3 consecutive financial years.
3. **Accrual Divergence:** CFO / PAT ratio < 0.50x over a 3-year average, signaling aggressive revenue recognition.

8. Statistical Clustering & ML Financial Archetypes

To discover hidden structures and groupings across the Nifty 100 universe, src/analytics/clustering.py applies unsupervised **K-Means Clustering ($k=5$)** on normalized multi-dimensional financial features.

Cluster ID & Name	Constituent Count	Defining Financial Signature & Profile
0: High-Quality Compounders	2 Companies	Exceptional ROE (>35%), negligible leverage (D/E < 0.1), strong pricing power and high FCF yields.
1: Defensive Dividend Payers	58 Companies	Mature market leaders, stable cash flows, consistent dividend yields, conservative balance sheets.
2: Value Cyclicals	2 Companies	Capital-intensive, commodity or credit cyclical exposure, low valuation multiples, volatile OPM.
3: Distressed / Turnaround	14 Companies	Depressed ROE, elevated gearing ratios (D/E > 2.0x), negative or highly volatile free cash flow.
4: Emerging Growth	16 Companies	Rapid top-line expansion (Revenue CAGR > 18%), aggressive CapEx reinvestment, moderate margins.

Mathematical Methodology & Normalisation

Feature Imputation & Scaling: Missing metrics are imputed using sector medians. All five input features (ROE %, Debt-to-Equity, Revenue CAGR 5Y, FCF CAGR 5Y, and OPM %) are scaled to zero mean and unit variance via StandardScaler. The optimal number of clusters ($k=5$) was confirmed using the inertia elbow plot (reports/elbow_plot.png).

9. Streamlit Dashboard: 8-Screen User Manual

The interactive web dashboard is launched via streamlit run src/dashboard/app.py and runs on http://localhost:8501. The sidebar allows zero-latency navigation across all 8 specialized analytical screens:

Screen ID & Name	Key Visual Widgets & Functionality	Data Source & Interactivity
01 - Home	Market summary tiles (Avg ROE, Med P/E, Med D/E), Plotly sector donut, Top 5 quality compounders.	Global year selector (2019 to 2024); instant recalculation.
02 - Company Profile	Company card, 6 KPI cards, 10Y Revenue/PAT dual bars, ROE/ROCE lines, automated NLP pros/cons badges.	Instant autocomplete search dropdown by Ticker or Name.
03 - Screener	10 metric sidebar sliders, 6 one-click preset buttons, live-updating filtered table, CSV download button.	Filters applied in-memory across 92 companies with instant response.
04 - Peer Comparison	Interactive 8-axis Plotly radar polygon overlaying company vs peer group average, side-by-side KPI table.	Dropdowns for 11 Peer Groups and constituent ticker selection.
05 - Trend Analysis	10-year time-series trend line charts with YoY growth rate percentage labels.	Multi-metric overlay: select up to 3 financial KPIs simultaneously.
06 - Sector Analysis	Scatter bubble chart (X=Revenue, Y=ROE, Bubble Size=Market Cap, Color=Sub-Sector), median KPI bar chart.	Macro sector selector (11 sectors) with sub-sector coloring.
07 - Capital Allocation	Interactive Plotly Treemap mapping 92 companies across the 8 cash flow allocation regimes.	Interactive zoom/drill-down into constituent companies.
08 - Annual Reports	Repository of direct BSE India annual report filing URLs with automated 404 validity checking.	Company search box with status badges (■ Available, ■ 404).

10. REST API Specification & Integration Guide

The REST API is built on FastAPI and Uvicorn, serving 16 endpoints prefixed with /api/v1. Interactive Swagger documentation is available at http://localhost:8000/docs.

HTTP Method & Route	Parameters & Filtering	Response Payload Description
GET /api/v1/health	None	Status 'ok', uptime_seconds, and exact row counts for all 10 SQLite tables.
GET /api/v1/companies	?sector=&market;_cap_category=&search;=	Array of 92 companies with metadata, latest ROE, and ROCE metrics.
GET /api/v1/companies/{ticker}	Path: ticker (e.g. TCS)	Full profile object: description, sector, latest ratios, and pros/cons.
GET /api/v1/companies/{ticker}/pl	?from_year=&to;_year=	Longitudinal P&L; history: sales, operating_profit, net_profit, EPS.
GET /api/v1/companies/{ticker}/bs	?from_year=&to;_year=	Balance Sheet series: equity, reserves, borrowings, total_assets.
GET /api/v1/companies/{ticker}/cashflow	?from_year=&to;_year=	Statutory cash flow series: CFO, CFI, CFF, net_cash_flow.
GET /api/v1/companies/{ticker}/ratios	?year=	Array of 14 computed KPIs per year including CAGRs and margins.
GET /api/v1/companies/{ticker}/tear sheet	Path: ticker	Direct binary PDF stream download of 2-page company tearsheet.
GET /api/v1/screener	?min_roe=&max;_de=&sector;=&min;_rev_c agr_5yr=	Filtered, ranked JSON array of matching companies with composite score.
GET /api/v1/sectors	None	Summary of 11 macro sectors with median ROE, P/E, and D/E benchmarks.
GET /api/v1/peers/{group_name}	Path: group_name (e.g. IT Services)	Percentile rank table across 8 axes for all member companies.
GET /api/v1/portfolio/stats	None	Portfolio distribution: P10, P25, P50, P75, P90, Mean, Std for 10 KPIs.

Sample cURL Invocation

```
curl -X GET 'http://localhost:8000/api/v1/screener?min_roe=15&max;_de=1.0' -H 'Accept: application/json'
```

11. Data Quality Rules & Production Runbook

The platform implements 16 automated Data Quality (DQ) validation rules executed during raw data ingestion. Violations are triaged and logged to output/validation_failures.csv:

Rule ID & Name	Validation Condition	Severity	Pipeline Enforcement Action
DQ-01: Company PK	companies.id unique (92 unique)	CRITICAL	Halt pipeline immediately on duplicate.
DQ-02: Annual PK	Unique (company_id, year) pair	CRITICAL	Deduplicate: keep latest occurrence.
DQ-03: FK Integrity	child.company_id in companies.id	CRITICAL	Reject orphan rows; log to CSV.
DQ-04: BS Match	|Assets - Liabilities| / Assets < 0.01	WARNING	Flag row for analyst review.
DQ-05: OPM Check	|OPM_calc - OPM_source| < 1.0%	WARNING	Use computed OPM; keep source for display.
DQ-06: Non-Zero Sales	Sales > 0 for non-banks	WARNING	Exclude zero sales rows from CAGR growth.
DQ-07: Year Format	Normalized matches r'^\d{4}-\d{2}$'	CRITICAL	Reject unparseable dates (e.g. TTM).
DQ-08: Ticker Format	2 ≤ length(ticker) ≤ 12, uppercase	CRITICAL	Normalize whitespace; reject out-of-range.
DQ-09: Cash Integrity	|NCF - (CFO+CFI+CFF)| ≤ 10 Cr	WARNING	Recompute NCF from components.
DQ-10: Fixed Assets	Fixed Assets ≥ 0	WARNING	Coerce negative values to 0.
DQ-11: Tax Rate Range	0% ≤ Tax Rate ≤ 60%	WARNING	Flag out-of-range tax reversals.
DQ-12: Payout Cap	Dividend Payout ≤ 200%	WARNING	Flag probable data-entry error.
DQ-13: URL Check	requests.head() status == 200	WARNING	Log 404 URLs without rejecting row.
DQ-14: EPS Sign	EPS > 0 if PAT > 0	WARNING	Flag dilution / accounting adjustment.
DQ-16: History Check	Company records ≥ 5 years	WARNING	Flag short tenure; exclude from CAGR.

Production Quick-Start Execution Sequence

```
# 1. Build SQLite Data Warehouse: python src/etl/loader.py
# 2. Compute 50+ Financial KPIs: python -m src.analytics.ratio_engine
# 3. Generate Screener & Scored: python -m src.screener.engine
# 4. Generate All 92 Tearsheets: python -m src.reports.tearsheet
# 5. Launch Web Dashboard: streamlit run src/dashboard/app.py
```